

1. Introduction

Artificial Neural Networks (ANNs) are computational models inspired by the structure and functioning of the human brain. They are designed to simulate the way biological neurons process information, learn from experience, and make decisions. ANNs form one of the fundamental pillars of modern Artificial Intelligence (AI) and Machine Learning (ML), enabling systems to recognize patterns, classify data, predict outcomes, and solve complex real-world problems.

The concept of neural networks originated from the early computational model of a neuron proposed by Warren McCulloch and Walter Pitts in 1943. Later, the development of the Perceptron by Frank Rosenblatt marked a significant milestone in neural network research. Over time, advancements in computing power, availability of large datasets, and improved learning algorithms have led to the emergence of deep learning models that significantly outperform traditional methods in many domains.

An Artificial Neural Network consists of interconnected processing units called *neurons* or *nodes*. These neurons are organized into layers: an input layer, one or more hidden layers, and an output layer. Each connection between neurons has an associated weight that determines the strength of the signal being passed. The network processes input data by applying weighted sums followed by nonlinear activation functions, producing outputs that represent predictions or classifications.

Today, ANNs are widely applied in various fields such as image recognition, speech processing, natural language understanding, medical diagnosis, financial forecasting, robotics, and autonomous systems. Modern deep learning architectures such as Convolutional Neural Network and Recurrent Neural Network have revolutionized industries by achieving human-level or even superhuman performance in specific tasks.

Passive Decay Model

$$\frac{dx_i(t)}{dt} = -A_i x_i(t)$$

Symbols

- $x_i(t)$: activation (state) of neuron indexed by i at time t
- i : index identifying the neuron
- t : time variable
- $\frac{dx_i(t)}{dt}$: derivative of $x_i(t)$ with respect to time (rate of change)
- A_i : positive decay (leakage) constant for neuron i
- $-$: indicates reduction (decay) of activation

Modified Passive Decay Model

$$\frac{dx_i(t)}{dt} = -\frac{A_i}{C_i} x_i(t)$$

Symbols

- $x_i(t)$: activation of neuron i at time t
- t : time

- $\frac{dx_i(t)}{dt}$: rate of change of activation
- A_i : decay constant
- C_i : membrane capacitance of neuron i
- $\frac{A_i}{C_i}$: effective decay rate
- $-$: decay effect

Non-Zero Resting Potential Model

$$\frac{dx_i(t)}{dt} = -A_i x_i(t) + P_i$$

Symbols

- $x_i(t)$: activation of neuron i at time t
- t : time
- $\frac{dx_i(t)}{dt}$: rate of change of activation
- A_i : decay constant
- P_i : constant input (resting potential or bias)
- $+$: additive effect

External Input Activation Model

$$\frac{dx_i(t)}{dt} = -A_i x_i(t) + B_i I_i$$

Symbols

- $x_i(t)$: activation of neuron i at time t
- t : time

- $\frac{dx_i(t)}{dt}$: rate of change of activation
- A_i : decay constant
- B_i : input gain (scaling factor)
- I_i : external input signal applied to neuron i
- $+$: additive contribution

Additive Autoassociative Model

$$\frac{dx_i(t)}{dt} = -A_i x_i(t) + B_i I_i + \sum_{j=1}^N W_{ij} f_j(x_j(t))$$

Symbols

- $x_i(t)$: activation of neuron i at time t
- t : time
- $\frac{dx_i(t)}{dt}$: rate of change of activation
- A_i : decay constant
- B_i : input gain
- I_i : external input to neuron i
- $\sum_{j=1}^N$: summation over index j from 1 to N
- j : index of presynaptic neuron
- N : total number of neurons
- W_{ij} : synaptic weight from neuron j to neuron i
- $f_j(\cdot)$: output function of neuron j
- $x_j(t)$: activation of neuron j at time t

Inhibitory Feedback Model

$$\frac{dx_i(t)}{dt} = -A_i x_i(t) - B_i I_i - \sum_{j=1}^N W_{ij} f_j(x_j(t))$$

Symbols

- $x_i(t)$: activation of neuron i at time t
- t : time
- $\frac{dx_i(t)}{dt}$: rate of change
- A_i : decay constant
- B_i : scaling constant
- I_i : external input
- $\sum_{j=1}^N$: summation over neurons
- j : neuron index
- N : total neurons
- W_{ij} : weight from neuron j to neuron i
- $f_j(x_j(t))$: output of neuron j

Perkel's Model

$$\frac{dx_i(t)}{dt} = \frac{1}{R_i} x_i(t) + \sum_{j=1}^N \frac{1}{R_{ij}} \phi_j(x_j(t))$$

Symbols

- $x_i(t)$: activation of neuron i
- t : time
- $\frac{dx_i(t)}{dt}$: rate of change
- R_i : resistance of neuron i
- $\frac{1}{R_i}$: conductance
- $\sum_{j=1}^N$: summation over neurons
- j : neuron index
- N : total neurons
- R_{ij} : resistance between neurons i and j
- $\phi_j(\cdot)$: output function
- $x_j(t)$: activation of neuron j

Hopfield Model

$$\frac{dx_i(t)}{dt} = -A_i x_i(t) + \sum_{j=1}^N w_{ij} f(x_j(t)) + I_i$$

Symbols

- $x_i(t)$: activation of neuron i
- t : time
- $\frac{dx_i(t)}{dt}$: rate of change
- A_i : decay constant
- $\sum_{j=1}^N$: summation

- j : neuron index
- N : total neurons
- W_{ij} : symmetric connection weight
- $f(\cdot)$: bounded activation function
- $x_j(t)$: activation of neuron j
- I_i : external input

Heteroassociative Network

Layer-1

$$\frac{dx_i(t)}{dt} = -A_i x_i(t) + \sum_{j=1}^M f_j(y_j(t)) V_{ji} + I_i$$

Symbols

- $x_i(t)$: activation of neuron i in layer-1
- t : time
- $\frac{dx_i(t)}{dt}$: rate of change
- A_i : decay constant
- $\sum_{j=1}^M$: summation over layer-2 neurons
- j : index of neuron in layer-2
- M : number of neurons in layer-2
- $f_j(\cdot)$: output function of neuron j
- $y_j(t)$: activation of neuron j in layer-2
- V_{ji} : weight from neuron j to neuron i
- I_i : external input

Layer-2

$$\frac{dy_j(t)}{dt} = -B_j y_j(t) + \sum_{i=1}^N g_i(x_i(t)) W_{ij} + J_j$$

Symbols

- $y_j(t)$: activation of neuron j in layer-2
- t : time
- $\frac{dy_j(t)}{dt}$: rate of change
- B_j : decay constant
- $\sum_{i=1}^N$: summation over layer-1 neurons
- i : neuron index in layer-1
- N : number of neurons in layer-1
- $g_i(\cdot)$: output function of neuron i
- $x_i(t)$: activation of neuron i
- W_{ij} : weight from neuron i to neuron j
- J_j : external input

Bidirectional Associative Memory (BAM)

Same equations as heteroassociative network with

- $V = W^T$: weight matrix V equals transpose of weight matrix W

Basic Shunting Model

$$\frac{dx_i(t)}{dt} = -A_i x_i(t) + (B_i - x_i(t))I_i$$

Symbols

- $x_i(t)$: activation
- t : time
- $\frac{dx_i(t)}{dt}$: rate of change
- A_i : decay constant
- B_i : upper saturation limit
- I_i : external input
- $(B_i - x_i(t))$: remaining activation capacity